THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING: LEVEL I EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL1105: Environmental Engineering

May 18, 2018 TIME: 2:30 pm – 5:30 pm

I. Answer ALL questions. Section A carries 30 marks, and Section B carries 70 marks.

II. Atomic weights: H 1; C 12; N 14; O 16; F 19; Na 23; Mg 24.3; Al 27; P 31; S 32; Cl 35.5; K 39; Ca 40; Cr 52; Mn 55; Fe 55.8; Ni 58.7; Cu 63.5; Zn 65.4; I 127.

III. Properties of water:

Density = 1000 kg/m^3
Viscosity (15 °C) = 1.1404 mPa·s
Specific heat = 4.18 kJ/(kg.°C)
Heat of vaporization (15 °C) = 2465 kJ/kg
Heat of fusion = 333 kJ/kg

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Section A. (10 multiple choice questions with 3 marks each)

1. Which process/devise is suitable for reducing NO_x emission from coal fired power plants? (a) Cyclone, (b) three-way catalytic convertor, (c) oxidation pond, (d) selective catalytic reduction, (e) electrostatic precipitator, or (f) none of the above.
2. Which type of halogenated compounds are greenhouse compounds but NOT ozone depleting compounds? (a) halons, (b) CFCs, (c) HCFCs, (d) HFCs, or (e) none of the above.
3. Which compound has longest life span in atmosphere? (a) CH_4, (b) CF_3Br, (c) CF_2Cl_2, (d) CF_4, or (e) CF_2HCl
4. How does adding alum (a common coagulant) into a water solution change its alkalinity (ALK)? (a) ALK increases, (b) ALK does not change, (c) ALK decreases, or (d) ALK decreases initially and then increases.
5. Colloidal particles in natural water are typically negatively charged. Which of the following actions does NOT negatively impact the stability of colloids? (a) Adding multivalent cations (such as Ca^{2+}) to the solution, (b) neutralization of charge of the colloidal particles, (c) adding positively charged polymers to the solution, or (d) none of the above.
6. Conventional wastewater treatment is NOT designed to remove (a) TSS, (b) TDS, (c) BOD, (d) pathogen, or (e) none of the above.
7. Which compound is NOT considered as a greenhouse compound? (a) CO_2, (b) SO_2, (c) N_2O, (d) O_3, or (e) none of the above.
8. Which biological treatment belongs to the attached growth type? (a) Activated sludge, (b) aerated lagoon, (c) trickling filter, (d) oxidation pond, or (e) none of the above.
9. For the production of ultrapure water, one often uses an RO treatment followed by an ion exchange treatment. A typical RO process can remove 99% ions from the feed water. In order to achieve an overall ion removal of 99.95%, what is the required removal efficient for the ion exchange process? (a) 0.5%, (b) 0.95%, (c) 5%, (d) 9.5%, (e) 50%, or (f) 95%.
10. Thermal stratification (a) usually occurs in the fall (autumn), (b) causes acidification of the water body, (c) occurs only in tropical regions, (d) is caused by high water temperature, or (e) none of the above.

Section B (Three questions with unequal marks)

11. (a) Briefly explain the following terminologies (**use < 80 words for each item**):
    i. Eutrophication;
    ii. Temperature inversion;
    iii. Membrane bioreactor;
    iv. Secondary air pollutants.

    (24 marks)

    (b) Use one specific example to illustrate positive feedback loop in the context of global warming.

    (6 marks)

12. (a) A wastewater contains 300 mg/L C_6H_7N. Determine its total oxygen demand and carbonaceous oxygen demand.

(8 marks)

(b) A sample is collected from this wastewater to perform a BOD test. Briefly explain what is BOD and how to measure BOD. Sketch the measured BOD value as a function of time for this wastewater sample.

(12 marks)

13. (a) The temperature at the ground level is 20 °C. The temperature decreases linearly to 15 °C at an elevation of 300 m and then it remains constant at 15 °C above 300 m. Determine the mixing depth.

(4 marks)

(b) A city has a rectangular shape with a width of 10 km and a length of 20 km. There are 100,000 cars driving in the city during the evening peak hours. On average, each car is assumed to drive at 50 km/hour, and it emits 0.2 g/km of NO_x. Use a box model to estimate the NO_x concentration in the city. You can assume the following:

- Mixing depth: 100 m
- Wind: 0.5 m/s (along the length of the city)
- NO_x concentration in the upstream air is negligible

(12 marks)

(c) List two environmental/health effects of NO_x.

(4 marks)

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